

Ravi Prakash Srivastava

📞 +91-9431334691 ✉️ raviprakashshrivastav7@gmail.com 🌐 ravi-prakash-sri 📄 er-ravi-srivastava.github.io

Research Interests

Large Language Models (LLMs), AI Agents, Trustworthy AI, Machine Learning Systems, Adversarial Machine Learning, Computer Vision.

Education

Indian Institute of Information Technology Ranchi (IIIT Ranchi), India August 2024 – May 2026
M.Tech (Data Science & AI) CGPA: 8.79/10

Institute of National Importance, Govt. of India

North Eastern Hill University (NEHU), Shillong, India August 2018 – June 2022
B.Tech (Information Technology) CGPA: 7.85/10

Central University, Govt. of India

Experience

Software Engineer *SS Software Solutions LLC, Hyderabad, India* March 2026 – Present

- Defined and enforced ML pipeline standards encompassing model versioning (MLflow/DVC), structured logging (JSON schema-validated with trace IDs for distributed tracing), and observability instrumentation (metrics, alerts, drift detection), reducing incident diagnosis time by 40% and improving pipeline reliability across 3+ production models.
- Standardized ML pipeline governance across versioning, logging, and monitoring layers, cutting untracked model deployments to zero and reducing mean-time-to-debug (MTTD) by ~35%; applied STRIDE-based Threat Modeling to identify 8+ design-level risks including data poisoning vectors and model inversion attack surfaces.
- Automated security testing by integrating SAST tools (Bandit, Semgrep) and dependency vulnerability scanning into CI/CD pipelines (GitHub Actions), reducing manual security review effort by ~60% and accelerating release cycles.

Machine Learning Intern *C-DAC (Govt. of India), Hyderabad, India* August 2022 – February 2023

- Performed large-scale data preprocessing and feature engineering on student activity logs (attendance, assignments, quizzes) to support predictive modeling.
- Implemented and benchmarked multiple machine learning models, including Decision Trees, Random Forests, and Support Vector Machines, for academic performance prediction.

Research Projects

- Adversarial Prompt Injection Attacks on LLM: Cryptographic Key Leakage and Defense Strategies**
Under Publication at International Conference on Advanced Computing and Applications (ICACA-2026).
- IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions using Deep Learning**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2025.2521688>
- Deepfake Video Detection Using Face-Centric Processing and Frame Sampling**
Under Publication International Conference on Advanced Computing and Applications (ICACA-2026).
- Balanced Few-Shot Episodic Learning for Accurate Retinal Disease Diagnosis**
(arXiv: <https://arxiv.org/abs/2512.04967v1>)
- Smart City Vehicle Accident Monitoring and Detection System Using MEMS, GSM, GPS Raspberry Pi 4**
[IETE Journal of Research] DOI: <https://doi.org/10.1080/03772063.2022.2043787>

Projects

AI-Powered Vulnerability Management Platform (SAST + DAST) Live Demo

- Automated SAST scanning using Semgrep for static code analysis and early vulnerability detection
- DAST testing covering SQLi, XSS, CSRF, and common web vulnerabilities in live environments

- Interactive Streamlit-based dashboard for real-time vulnerability visualization and reporting
- Vulnerability prioritization engine with security remediation recommendations
- CI/CD and DevSecOps-friendly architecture supporting seamless pipeline integration
- Dockerized deployment for scalability, portability, and consistent environment management

Technical Skills

- **Programming Languages:** Python, C++, SQL, Bash.
- **Machine Learning & AI:** Deep Learning, Machine Learning, Computer Vision, Image Processing, Deepfake Detection, Large Language Models (LLMs), Hugging Face Transformers, PyTorch, Scikit-learn, OpenCV.
- **MLOps & DevOps:** Git, Docker, GitHub Actions (CI/CD), MLflow, DVC.
- **Cybersecurity & Trustworthy AI:** Adversarial Machine Learning, LLM Prompt Injection, Threat Modeling (STRIDE), OWASP, Vulnerability Assessment, Penetration Testing, SAST (Bandit, Semgrep), IDS/IPS.

Achievements

- Received the **Institution of Electronics and Telecommunication Engineers (IETE) Scholarship** for academic excellence during M.Tech. studies in Computer Science and Engineering.
- Received the **Best Paper Award** at **ICACA 2026** for the paper, *"Adversarial Prompt Injection Attacks on Large Language Models: Cryptographic Key Leakage and Defense Strategies."*